

NATIONAL BOARD OF ACCREDITATION

Data Capturing Points of the Program Applied for NBA Accreditation– Tier I/II UG (Engineering) Institute Programs

Program Name : Computer Science and Engineering	Discipline : Engineering & Technology
Level : Under Graduate	Tier : 1
Application No : 11860	Date of Submission : 20-05-2026

PART A- Profile of the Institute

A1.Name of the Institute: KLE Technological University	
Year of Establishment : 1947	Location of the Institute: BVB Campus, Vidyanagar, Hubballi
A2. Institute Address: KLE Technological University B.V. Bhoomaraddi College Campus , Vidyanagar Hubballi -580031	
City:Dharwad	State:Karnataka
Pin Code:580031	Website:www.kletech.ac.in
Email:vc@kletech.ac.in	Phone No(with STD Code):0836-2378102
A3. Name and Address of the Affiliating University (if any):	
Name of the University :	City:
State :	Pin Code: 0
A4. Type of the Institution: University	
A5. Ownership Status: Self financing	

A6. Details of all Programs being Offered by the Institution:

- No. of UG programs: 11
- No. of PG programs: 8

Table No. A6.1: List of all programs offered by the Institute.

Sr.No.	Discipline	Level of program	Name of the program	Year of Start	Year of Closed	Name of The Department
1	Architecture	UG	Architecture	2015	--	Architecture
2	Computer Application	PG	Master of Computer Application	2015	--	Computer Application
3	Computer Application	PG	Master of Computer Applications (Integrated)	2024	--	Computer Application
4	Engineering & Technology	PG	Advanced Manufacturing Systems	2020	--	Mechanical Engineering
5	Engineering & Technology	UG	Automation & Robotics	2015	--	Automation and Robotics
6	Engineering & Technology	UG	Biotechnology	2015	--	Biotechnology
7	Engineering & Technology	UG	Civil Engineering	2015	--	Civil Engineering
8	Engineering & Technology	UG	Computer Science and Engineering	2015	--	Computer Science and Engineering
9	Engineering & Technology	PG	Computer Science and Engineering	2015	--	Computer Science and Engineering

10	Engineering & Technology	UG	Computer Science and Engineering (Artificial Intelligence)	2021	--	Computer Science and Engineering (Artificial Intelligence)
11	Engineering & Technology	PG	Design Engineering	2020	--	Mechanical Engineering
12	Engineering & Technology	UG	Electrical & Electronics Engineering	2015	--	Electrical and Electronics Engineering
13	Engineering & Technology	UG	ELECTRONICS AND COMMUNICATION ENGINEERING	2015	--	Electronics and Communication Engineering
14	Engineering & Technology	UG	Electronics and Communication Engineering (Industry Integrated)	2020	--	Electronics and Communication Engineering
15	Engineering & Technology	UG	Electronics Engineering (VLSI Design and Technology)	2022	--	Electronics Engineering (VLSI Design and Technology)
16	Engineering & Technology	UG	Mechanical Engineering	2015	--	Mechanical Engineering
17	Engineering & Technology	PG	Structural Engineering	2015	--	Civil Engineering
18	Engineering & Technology	PG	VLSI Design & Embedded Systems	2015	--	Electronics and Communication Engineering
19	Management	PG	Master of Business Administration	2015	--	Management

A7. Programs to be considered for Accreditation vide this Application:

Table No. A7.1: List of programs to be considered for accreditation.

Name of the Department	Having Allied Departments	Name of the Program	Program Level
Biotechnology	No	Biotechnology	UG
Electronics and Communication Engineering	Yes	ELECTRONICS AND COMMUNICATION ENGINEERING	UG
Computer Science and Engineering	Yes	Computer Science and Engineering	UG

Table No. A7.2: Allied Department(s) to the Department of the program considered for accreditation as above.

Cluster ID: Name of the Department (in table no. A7.1) Name of allied Departments/Cluster (for table no. A7.1)

Allied Department/Cluster Name	Program Name	Program Level
Computer Science and Engineering (Artificial Intelligence)	Computer Science and Engineering (Artificial Intelligence)	UG

PART-B: Program information**B1. Provide the Required Information for the Program Applied For:**

Table No. B1: Program details.

A. List of the Programs Offered by the Department:

SR.NO.	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITED	PROGRAM DURATION
1	Computer Science and Engineering	UG	2015 / --	300	Yes	2025	480	2025	F. No. South-West/1-44638523851/2025/EOA 21-Mar-2025	Granted accreditation for 3 years for the period (specify period)	2023	2026	2	4

Sanctioned Intake for Last Five Years for the Computer Science and Engineering	
Academic Year	Sanctioned Intake
2025-26	480
2024-25	300
2023-24	300
2022-23	300
2021-22	300
2020-21	300

List of the Allied Departments/Cluster and Programs:

SR.NO.	ALLIED DEPARTMENT NAME	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITE
1	Computer Science and Engineering (Artificial Intelligence)	Computer Science and Engineering (Artificial Intelligence)	UG	2021 / --	60	Yes	2023	120	2023	F. No. South-West/1-44638523851/2025/EOA 21-Mar-2025	Not eligible for accreditation	--	--	0

Sanctioned Intake for Last Five Years for the Computer Science and Engineering (Artificial Intelligence)	
Academic Year	Sanctioned Intake
2025-26	120
2024-25	120
2023-24	120
2022-23	60
2021-22	60
2020-21	0

B2. Detail of Head of the Department for the program under consideration:

A. Name of the HoD :	Dr. Vijayalakshmi M.
B. Nature of appointment:	Regular
C. Qualification:	Ph.D

B3. Program Details

Table No.B3.1: Admission details for the program excluding those admitted through multiple entry and exit points.

Item (Information to be provided cumulatively for all the shifts with explicit headings, wherever applicable)	2025-26 (CAY)	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)	2021-22 (CAYm4)	2020-21 (CAYm5)	2019-20 (CAYm6)
N=Sanctioned intake of the program (as per AICTE /Competent authority)	480	300	300	300	300	300	300
N1=Total no. of students admitted in the 1st year minus the no. of students, who migrated to other programs/ institutions plus no. of students, who migrated to this program	464	294	307	307	319	303	303
N2=Number of students admitted in 2nd year in the same batch via lateral entry including leftover seats	0	35	28	30	32	30	27
N3=Separate division if any	0	0	0	0	0	0	0
N4=Total no. of students admitted in the 1st year via all supernumerary quotas	0	0	0	0	0	0	0
Total number of students admitted in the program (N1 + N2 + N3 + N4) - excluding those admitted through multiple entry and exit points.	464	329	335	337	351	333	330

CAY= Current Academic Year. CAYm1= Current Academic Year Minus 1 CAYm2= Current Academic Year Minus 2. LYG= Last Year Graduate. LYGm1= Last Year Graduate Minus 1. LYGm2= Last Year Graduate Minus 2.

B4. Enrolment Ratio in the First Year

Table No. B4.1: Student enrolment ratio in the 1st year.

Year of entry	N (From Table 4.1)	N1 (From Table 4.1)	N4 (From Table 4.1)	Enrollment Ratio [(N1/N)*100]
2025-26 (CAY)	480	464	0	96.67
2024-25 (CAYm1)	300	294	0	98.00
2023-24 (CAYm2)	300	307	0	102.33

Average [(ER1 + ER2 + ER3) / 3] = 99.00 $\pm$ 20.00**B5. Success Rate of the Students in the Stipulated Period of the Program**

Table No.B5.1: The success rate in the stipulated period of a program.

Item	(2021-22) LYG	(2020-21) LYGm1	(2019-20) LYGm2
A*= (No. of students admitted in the 1st year of that batch and those actually admitted in the 2nd year via lateral entry, plus the number of students admitted through multiple entry (if any) and separate division if applicable, minus the number of students who exited through multiple entry (if any).	351.00	333.00	330.00
B=No. of students who graduated from the program in the stipulated course duration	318.00	311.00	320.00

Average SR of three batches ((SR_1+ SR_2+ SR_3)/3): 93.65

B6. Academic Performance of the First-Year Students of the Program

Table No.B6.1: Academic Performance of the First-Year Students of the Program.

Academic Performance	CAYm1(2024-25)	CAYm2(2023-24)	CAYm3 (2022-23)
Mean of CGPA or mean percentage of all successful students(X)	8.22	8.20	8.22
Y=Total no. of successful students	276.00	295.00	307.00
Z=Total no. of students appeared in the examination	291.00	302.00	311.00
API [X*(Y/Z)]	7.80	8.01	8.11

Average API[(AP1+AP2+AP3)/3] : 7.97

B7: Academic Performance of the Second Year Students of the Program

Table No.B7.1: Academic Performance of the Second Year Students of the Program.

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 2nd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 2nd year/10)	8.10	8.03	7.94
Y=Total no. of successful students	308.00	331.00	328.00
Z=Total no. of students appeared in the examination	323.00	337.00	336.00
API [X * (Y/Z)]	7.72	7.89	7.75

Average API [(AP1 + AP2 + AP3)/3] : 7.79

B8. Academic Performance of the Third Year Students of the Program

Table No.B8.1: Academic Performance of the Third Year Students of the Program

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 3rd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 3rd year/10)	8.20	8.18	8.03
Y=Total no. of successful students	329.00	326.00	316.00
Z=Total no. of students appeared in the examination	331.00	328.00	316.00
API [X*(Y/Z)]:	8.15	8.13	8.03

Average API [(AP1 + AP2 + AP3)/3] : 8.10

B9. Placement, Higher Studies, and Entrepreneurship

Table No.B9.1: Placement, higher studies, and entrepreneurship details.

Item	LYG (2021-22)	LYGm1(2020-21)	LYGm2(2019-20)
FS*=Total no. of final year students	333.00	330.00	327.00
X=No. of students placed	234.00	209.00	255.00
Y=No. of students admitted to higher studies	10.00	6.00	20.00
Z= No. of students taking up entrepreneurship	1.00	0.00	0.00
Placement Index(P) = (((X + Y + Z)/FS) * 100):	73.57	65.15	84.10

Average Placement Index = (P_1 + P_2 + P_3)/3: 74.27 Placement Index Points:

PART C: Faculty Details in Department and Allied Departments**(Data to be filled in for the Department and Allied Departments)**

C1. Faculty details of Department and Allied Departments

Table No.C1: Faculty details in the Department for the past 3 years including CAY

Sr.No	Name of the Faculty	PAN No.	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?
1	Dr. Meena S. Maralappanavar	XXXXXX67C	Ph.D	Visvesvaraya Technological University, Belagavi	Image Processing, Information Retrieval, Machine Learning	08/05/1989	37	Lecturer	Professor	18/04/2012	Contractual Fulltime	Yes		No
2	Dr. Vijayalakshmi M.	XXXXXX99H	Ph.D	Jawaharlal Nehru Technological University, Hyderabad	Wireless Multimedia Networks	19/01/2009	17.3	Assistant Professor	Professor	02/08/2021	Regular	Yes		Yes
3	Dr. Satyadhyam Chickerur	XXXXXX37P	Ph.D	Visvesvaraya Technological University, Belagavi	HPC, Deep Learning, GPU Programming	02/01/2013	13.4	Professor	Professor	02/01/2013	Regular	Yes		No
4	Dr. Karibasappa K.G.	XXXXXX31F	Ph.D	Jawaharlal Nehru Technological University, Hyderabad	Data Analytics	02/09/2002	23.8	Lecturer	Professor	01/04/2017	Regular	Yes		No
5	Dr. Shrinivas D. Desai	XXXXXX62C	Ph.D	Visvesvaraya Technological University, Belagavi	Medical Image Processing	20/05/2000	25.11	Lecturer	Professor	02/08/2021	Regular	Yes		No
6	Dr. Sujata C	XXXXXX95J	Ph.D	Visvesvaraya Technological University, Belagavi	Image and Video processing	05/07/2002	23.10	Lecturer	Professor	02/08/2021	Regular	Yes		No
7	Dr. Suvarna .G. Kanakaraddi	XXXXXX31P	Ph.D	Visvesvaraya Technological University, Belagavi	Artificial Intelligence, Data Science	27/02/2003	23.2	Lecturer	Professor	02/08/2021	Regular	Yes		No
8	Dr. P.G. Sunitha Hiremath	XXXXXX05N	Ph.D	Jawaharlal Nehru Technological University, Hyderabad	Computer Networks, Data Analytics	16/07/2010	15.9	Assistant Professor	Professor	02/08/2021	Contractual Fulltime	Yes		No
9	Dr. Jayalaxmi G.N.	XXXXXX28G	Ph.D	Visvesvaraya Technological University, Belagavi	Wireless Mesh Network	08/09/2001	24.8	Lecturer	Professor	01/03/2023	Regular	Yes		No

10	Dr. Padmashree Desai	XXXXXX63N	Ph.D	Jawaharlal Nehru Technological University, Hyderabad	Image Processing	16/08/2007	18.9	Assistant Professor	Professor	01/03/2023	Regular	Yes		No
11	Dr. Manohar Madgi	XXXXXX70B	Ph.D	Jawaharlal Nehru Technological University, Hyderabad	Image Processing	01/09/2023	2.5	Professor	Professor		Regular	No	28/02/2026	No
12	Dr. Vishwanath P. Baligar	XXXXXX88R	Ph.D	Indian Institute of Science, Bangalore	Image Processing	04/06/2012	13.11	Professor	Professor		Contractual Fulltime	Yes		No
13	Dr Shrinivas K. Kulkarni	XXXXXX14Q	Ph.D	Geneva, SSBS University of Edinburgh	Cyber Security	05/08/2024	1.9	Professor	Professor		Contractual Parttime	Yes		No
14	Dr. Gururaj S. Hanchinamani	XXXXXX93H	Ph.D	Visvesvaraya Technological University, Belagavi	Information Security	08/10/1993	32.7	Lecturer	Associate Professor	01/11/2011	Regular	Yes		No
15	Dr. Meenaxi M Raikar	XXXXXX25P	Ph.D	Visvesvaraya Technological University, Belagavi	Computer Networks	14/08/2009	16.9	Assistant Professor	Associate Professor	01/11/2011	Regular	Yes		No
16	Dr. Shantala Giraddi	XXXXXX19M	Ph.D	Visvesvaraya Technological University, Belagavi	Image Processing	21/01/2008	18.3	Lecturer	Associate Professor	01/03/2023	Contractual Fulltime	Yes		No
17	Dr. Sharada K. Shiragudikar	XXXXXX18J	Ph.D	Visvesvaraya Technological University, Belagavi	Deep learning	01/08/2024	1.6	Associate Professor	Associate Professor		Regular	No	28/02/2026	No
18	Dr. Ashok Chikaraddi	XXXXXX64F	Ph.D	Visvesvaraya Technological University, Belagavi	Wireless Multimedia Networks	04/08/2006	19.9	Lecturer	Associate Professor	01/03/2024	Regular	Yes		No
19	Mr. Chandarashekhar D. Kerure	XXXXXX08E	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	14/09/2001	24.8	Lecturer	Assistant Professor		Regular	Yes		No
20	Ms. Lalita Madanbhavi	XXXXXX05B	M.E.	Thapar Institute of Technology, Pathiala	Computer Science and Engineering	24/09/1994	31.7	Lecturer	Assistant Professor		Regular	Yes		No
21	Mr. K M M Rajashekaraiah	XXXXXX08F	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	21/06/2012	13.10	Assistant Professor	Assistant Professor		Regular	Yes		No

22	Dr. Guruprasad Konnurmath	XXXXXXXX29G	Ph.D	KLE Technological University, Hubballi	High Performance Computing	09/04/2021	5.1	Assistant Professor	Assistant Professor		Regular	Yes		No
23	Dr. Krupashankari Suresh Sandyal	XXXXXXXX76B	Ph.D	Visvesvaraya Technological University, Belagavi	Image Processing	16/03/2026	0.1	Assistant Professor	Assistant Professor		Regular	Yes		No
24	Mr. Manjunath K. Gonal	XXXXXXXX81D	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	31/10/2006	19.6	Lecturer	Assistant Professor		Regular	Yes		No
25	Mr. Parikshit P. Hegde	XXXXXXXX03R	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	25/05/2005	20.11	Lecturer	Assistant Professor		Regular	Yes		No
26	Mr. Anand S Meti	XXXXXXXX93N	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	15/07/2010	15.10	Lecturer	Assistant Professor		Regular	Yes		No
27	Ms. Nirmala Patil	XXXXXXXX16M	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	09/08/2010	15.9	Lecturer	Assistant Professor		Regular	Yes		No
28	Mr. Prakash Hegade	XXXXXXXX37H	M.Tech	IIIT- Bangalore	Computer Science and Engineering	30/07/2008	17.9	Lecturer	Assistant Professor		Regular	Yes		No
29	Ms. Indira Bidari	XXXXXXXX08K	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/08/2011	14.9	Lecturer	Assistant Professor		Regular	Yes		No
30	Mr. Sunil V. G.	XXXXXXXX51P	M.Tech	National Institute of Technology Karnataka, Surathkal	Computer Science and Engineering	01/08/2013	12.9	Assistant Professor	Assistant Professor		Regular	Yes		No
31	Ms. Manjula Pawar	XXXXXXXX40J	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	27/09/2012	13.7	Lecturer	Assistant Professor		Regular	Yes		No
32	Ms. Priyadarshini Patil	XXXXXXXX62M	M.Tech	Visvesvaraya Technological University, Belagavi	Digital Electronics	23/07/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No
33	Mr. Prashant M. Narayankar	XXXXXXXX86N	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	23/07/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No

34	Ms. Neha Tarannum	XXXXXXX42R	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/08/2011	14.9	Lecturer	Assistant Professor		Regular	Yes		No
35	Ms. N. V. Yaligar	XXXXXXX13J	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/09/2007	18.8	Lecturer	Assistant Professor		Regular	Yes		No
36	Ms. Vijayalakshmi Sajjanar	XXXXXXX05H	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	07/11/2022	3.6	Assistant Professor	Assistant Professor		Regular	Yes		No
37	Ms. Preethi Y R	XXXXXXX24D	M.Tech	Shri Bhagwan Mahaveer Jain College, Kanakpura Road, Bangalore/Jain University.	Computer Science and Engineering	10/08/2023	2.9	Assistant Professor	Assistant Professor		Regular	Yes		No
38	Ms. Uma M Hiremath	XXXXXXX54Q	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	01/08/2023	2.9	Assistant Professor	Assistant Professor		Regular	Yes		No
39	Mrs. Suma S. Barge	XXXXXXX46D	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	19/03/2024	2.1	Assistant Professor	Assistant Professor		Regular	Yes		No
40	Ms. Sweta Patil	XXXXXXX61F	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	25/07/2024	1.9	Assistant Professor	Assistant Professor		Regular	Yes		No
41	Ms. Sudha Tuppad	XXXXXXX73E	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	08/10/2024	1.7	Assistant Professor	Assistant Professor		Regular	Yes		No
42	Ms. Neha Panchal	XXXXXXX10E	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	16/10/2024	1.6	Assistant Professor	Assistant Professor		Regular	Yes		No
43	Ms. Muskan Indikar	XXXXXXX04H	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	07/10/2024	1.7	Assistant Professor	Assistant Professor		Regular	Yes		No
44	Ms. Vidya S. Handur	XXXXXXX43P	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	24/03/1998	28.1	Lecturer	Associate Professor	01/11/2011	Regular	Yes		No

45	Ms. Aruna S. Nayak	XXXXXX50D	M.Tech	Visvesvaraya Technological University, Belagavi	Digital Electronics	10/08/2007	18.9	Assistant Professor	Associate Professor	01/11/2011	Contractual Fulltime	Yes		No
46	Ms. Priyadarshini D Kalwad	XXXXXX44H	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/08/2005	20.9	Lecturer	Assistant Professor		Regular	Yes		No
47	Mr. Vijay S. Biradar	XXXXXX23C	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	09/02/2005	21.3	Lecturer	Assistant Professor		Regular	Yes		No
48	Ms. Kavitha H. S	XXXXXX35L	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	31/07/2008	17.9	Lecturer	Assistant Professor		Regular	Yes		No
49	Mr. Shivalingappa Battur	XXXXXX33A	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	02/09/2013	12.8	Assistant Professor	Assistant Professor		Regular	Yes		No
50	Mr. Sandeep Kulkarni	XXXXXX94N	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	16/01/2020	6.3	Assistant Professor	Assistant Professor		Regular	Yes		No
51	Ms. Swetha Kulkarni	XXXXXX67L	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	15/12/2022	3.4	Assistant Professor	Assistant Professor		Regular	Yes		No
52	Mrs Arati Joshi	XXXXXX08Q	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	07/08/2023	2.9	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
53	Ms Maheshwari M Kittur	XXXXXX91M	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	17/01/2023	3.3	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
54	Mrs Sarala D	XXXXXX49R	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	16/01/2023	3.3	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
55	Mrs Roopa Badami	XXXXXX08Q	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/01/2024	2.4	Assistant Professor	Assistant Professor		Regular	Yes		No

56	Mrs Swati M. Sajjan	XXXXXXX21R	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/9/2025	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
57	Ms Shilpa Hosagoudra	XXXXXXX67C	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	03/09/2025	0.6	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
58	Mr Pradeep Shet	XXXXXXX02P	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	01/12/2025	0.5	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
59	Shrushti Gunjal	XXXXXXX28N	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	01/12/2025	0.5	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
60	Sumayya Kadampur	XXXXXXX62H	M.Tech	KLE Technological University, Hubballi	Computer Science and Engineering	01/12/2025	0.5	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
61	Dr. Sumaiya Pathan	XXXXXXX18J	Ph.D	Manipal Academy of Higher Education, Manipal	Image Processing	17/09/2024	1.7	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
62	Ashrafunnissa A Mulla	XXXXXXX64H	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	05/12/2025	0.5	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
63	Mr. Praveen M Dhulavvagol	XXXXXXX16K	Ph.D	KLE Technological University	Big Data Analytics	15/07/2010	15.1	Lecturer	Assistant Professor		Regular	No	30/08/2025	No
64	Ms. Mradula Sharma	XXXXXXX91F	M.E.	Shri Venkateshwa University, Gajraula	computer science and engineering	01/08/2023	1.6	Assistant Professor	Assistant Professor		Regular	No	03/02/2025	No
65	Mr. Channabasappa Muttal	XXXXXXX53L	MS	National Research University- Higher School of Economics Moscow (Russia)	Computer Science and Engineering	15/04/2024	2	Assistant Professor	Assistant Professor		Regular	Yes		No
66	Mrs. Pratibha Malagatti	XXXXXXX52L	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	01/08/2023	2.9	Assistant Professor	Assistant Professor		Regular	Yes	15/02/2025	No

Table No.C2: Faculty details of Allied Departments for the past 3 years including CAY.

Sr.No	Name of the Faculty	PAN No.	APAAR faculty ID*(if any)	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?
1	Dr. Narayan D.G.	XXXXXXX13L	NA	Ph.D	Visvesvaraya Technological University, Belagavi	Computer Networks	18/11/2000	25.5	Assistant Professor	Professor	01/01/2020	Regular	Yes		Yes
2	Dr. Uday Kulkarni	XXXXXXX67M	NA	Ph.D	KLE Technological University	Machine Learning and AI, Deep Learning	06/01/2024	2.4	Associate Professor	Associate Professor		Regular	Yes		No
3	Ms. N D Kulenavar	XXXXXXX96Q	NA	M.Tech	Visvesvaraya Technological University, Belagavi		16/01/2001	25.4	Assistant Professor	Associate Professor	01/11/2011	Regular	Yes		No
4	Mr. Mallikarjun Akki	XXXXXXX37E	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	23/07/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No
5	Ms. Uma Devi F M	XXXXXXX72L	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	01/12/2010	15.5	Assistant Professor	Assistant Professor		Regular	Yes		No
6	Ms. Pooja Shettar	XXXXXXX10C	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	23/07/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No
7	Ms. Sneha Varur		NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	18/01/2021	5.3	Assistant Professor	Assistant Professor		Regular	Yes		No
8	Mr. Maheshgouda Patil	XXXXXXX57K	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	01/08/2023	2.9	Assistant Professor	Assistant Professor		Regular	Yes		No
9	Ms. Sadaf Anjum Savanur	XXXXXXX41F	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	25/10/2023	2.6	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No

10	Ms.Jyothi M Patil	XXXXXXXX85M	NA	M.Tech	Visvesvaraya Technological University, Belagavi	Networking and internet Engineering	30/08/2024	1.8	Assistant Professor	Assistant Professor		Regular	Yes		No
11	Ms. Arundati Aralimatti	XXXXXXXX47N	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	29/04/2024	2	Assistant Professor	Assistant Professor		Regular	Yes		No
12	Mr Amit Kachavimath	XXXXXXXX60N	NA	M.Tech	Visvesvaraya Technological University, Belagavi	computer science and engineering	31/12/2013	12.4	Assistant Professor	Assistant Professor		Regular	Yes		No
13	Dheeraj Kumar	XXXXXXXX60L	NA	M.Tech	KLE Technological University	computer science and engineering	07/11/2025	0.6	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No

C2. Student-Faculty Ratio (SFR)

No. of UG(Engineering) programs in Department including allied departments/ clusters (UGn):

UG1=1st UG program

UGn=nth UG program

B= No. of Students in UG 2nd year (ST)

C= No. of Students in UG 3rd year (ST)

D= No. of Students in UG 4th year (ST)

No. of PG (Engineering) programs in Department including allied departments/ clusters (PGm):

PG1=1st PG program.

PGm=mth PG program

A= No. of Students in PG 1st year

B= No. of Students in PG 2nd year

Student Faculty Ratio (**SFR**) = S/F

S= No. of students of all programs in the Department including all students of allied departments/clusters.

No. of students (ST)=Sanctioned Intake (SA)+ Actual admitted students via lateral entry including leftover seats (L) if any (limited to 10 % of SA)

Students who admitted under supernumerary quotas (SNQ, EWS, etc) will not be considered in calculating SFR value. Those students are exempted.

F=Total no. of regular or contractual faculty members (Full Time) in the Department, including allied departments/clusters (excluding first year faculty (The faculty members who have a 100% teaching load in the first-year courses)).

No. of UG Programs in the Department2 No. of PG Programs in the Department1

Table No.C2.1: Student-faculty ratio.

Description	CAY(2025-26)	CAYm1 (2024-25)	CAYm2 (2023-24)
UG1.B	330	328	330
UG1.C	322	327	330
UG1.D	326	330	329
UG1: Computer Science and Engineering	978	985	989
UG2.B	132	129	66
UG2.C	126	66	66
UG2.D	66	66	0
UG2: Computer Science and Engineering (Artificial Intelligence)	324	261	132

Description	CAY(2025-26)	CAYm1 (2024-25)	CAYm2 (2023-24)
PG1.A	18	18	24
PG1.B	18	24	24
PG1: Computer Science and Engineering	36	42	48
DS=Total no. of students in all UG and PG programs in the Department	1014	1027	1037
AS=Total no. of students of all UG and PG programs in allied departments	324	261	132
S=Total no. of students in the Department (DS) and allied departments (AS)	S1= 1338	S2= 1288	S3= 1169
DF=Total no. of faculty members in the Department	54	53	47
AF= Total no. of faculty members in the allied Departments	12	12	8
F=Total no. of faculty members in the Department (DF) and allied Departments (AF)	F1= 66	F2= 65	F3= 55
FF=The faculty members in F who have a 100% teaching load in the first-year courses	8	8	6
Student Faculty Ratio (SFR)=S/(F-FF)	SFR1= 23.07	SFR2= 22.60	SFR3= 23.86
Average SFR for 3 years	SFR= 23.18		

C3. Faculty Qualification

- Faculty qualification index (FQI) = 2.5 * [(10X +4Y)/RF] where
- X=No. of faculty members with Ph.D. degree or equivalent as per AICTE/UGC norms.
- Y=No. of faculty members with M. Tech. or ME degree or equivalent as per AICTE/ UGC norms.
- RF=No. of required faculty in the Department including allied Departments to adhere to the 20:1 Student-Faculty ratio, with calculations based on both student numbers and faculty requirements as per section C2 of this documents: (RF=S/20).

Table No.C3.1: Faculty qualification.

Year	X	Y	RF	FQ = 2.5 x [(10X + 4Y) / RF]]
2025-26(CAY)	19	47	66.00	14.32
2024-25(CAYm1)	19	46	64.00	14.61
2023-24(CAYm2)	16	39	58.00	13.62

C4. Faculty Cadre Proportion

- Faculty Cadre Proportion is 1(RF1): 2(RF2): 6(RF3)
- RF1= No. of Professors required = 1/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per C2 of this documents:.
- RF2= No. of Associate Professors required = 2/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- RF3= No. of Assistant Professors required = 6/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- Faculty cadre and qualification and experience should be as per AICTE/UGC norms.

Table No.C4.1: Faculty cadre proportion details.

Year	Professors		Associate Professors		Assistant Professors	
	Required RF1	Available AF1	Required RF2	Available AF1	Required RF3	Available AF3
2025-26	7.00	9.00	14.00	4.00	44.00	43.00
2024-25	7.00	10.00	14.00	5.00	42.00	41.00

2023-24	6.00	9.00	12.00	2.00	38.00	36.00
Average	RF1=6.67	AF1=9.33	RF2=13.33	AF2=3.67	RF2=41.33	AF2=40.00

C5. Visiting/Adjunct Faculty/Professor of Practice

Table No. C5.1: List of visiting/adjunct faculty/professor of practice and their teaching and practical loads.

(CAYm1)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	Dr. Shrinivas Kulkarni	Professor	KLE technological University	Cyber Security, Security Operations	60.00

(CAYm2)

(CAYm3)

C6. Academic Research

Table No. C6.1: Faculty publication details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
1	No. of peer reviewed journal papers published	39	36	20
2	No. of peer reviewed conference papers published	81	105	82
3	No. of books/book chapters published	3	6	6

C7. Sponsored Research Project

Table No. C7.1: List of sponsored research projects received from external agencies.

(CAYm1)

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Uma M	Dr. Meena S. M., Dr. Sujatha C	E&C, CSE	Digital Creation to Enable Gamification Based on Indian Ethos	SERB-INAE	2 yrs	63.89
						Amount received (Rs.):63.89

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Satyadhyan Chickerur		CSE	Meity-Qcal-QSDL-Diabetic Retinopathy (Meity)	MeiTY	2 yr	15.88
						Amount received (Rs.):15.88

Total Amount (Lacs) Received for the Past 3 Years: 79.77

Note*:

- Only sponsored research projects will be considered. Infrastructure-based projects will not be considered here.

C8. Consultancy Work

Table No. C8.1: List of consultancy projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Uma M	Dr. Meena S. M.	CSE	Samsung SEED Lab	Samsung SRIB	perpetual	70.12
						Amount received (Rs.):70.12

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Uma M	Dr. Meena S. M.	CSE	Samsung SEED Lab	Samsung SRIB	perpetual	77.82
						Amount received (Rs.):77.82

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Uma M	Dr. Meena S. M.	CSE	Samsung SEED Lab	Samsung SRIB	perpetual	10.96
						Amount received (Rs.):10.96

Total amount (Lacs) received for the past 3 years: 158.90

Note*:

- Only consultancy projects will be considered. Infrastructure-based projects will not be considered here.

C9. Institution Seed Money or Internal Research Grant to its Faculty for Research Work

Table No. C9.1: List of faculty members received seed money or internal research grant from the Institution.

(CAYm1)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. Satyadhyhan C (Director-CAIR)	CAIR	1 yr	8.50	8.50	60 Research Publications, 500 Student Engagement, 5 Applications for Ext Grants 3 Ph.D Awarded (UK. MP, GK)
			Amount received (Rs.): 8.50		

(CAYm2)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. Satyadhyhan C (Director- CAIR)	CAIR	1 yr	5.50	4.31	45 Research Publications, 400 Student engagement, 1 Ext Grant sanctioned
Prof. Umadevi F M Dr. Sujatha C	Indian Sign Language recognition	1 yr	1.00	1.00	Paper Publication - 02
			Amount received (Rs.): 6.50		

(CAYm3)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. Jayalaxmi Naragund	Resource management of mobile / multi access edge computing enabled 5G IoT N/w	1 yr	3.00	0.00	01 Scopus indexed Publication
Prof. Somashekhar Patil	Software Defined Network Testbed using Juniper Contrail	1 yr	16.00	0.00	01 Scopus indexed Publication
Dr. Vijeth Kotagi	IOT N/w and its implications in 5G N/w	1yr	3.00	0.00	01 Scopus indexed Publication
			Amount received (Rs.): 22.00		

Total amount (Lacs) received for the past 3 years : 37.00

PART D: Laboratory Infrastructure in the Department

(Data to be filled in for the Department)

D1. Adequate and Well-Equipped Laboratories, and Technical Manpower

Table No.D1.1: List of laboratories and technical manpower.

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Laboratory- 01	40	Desktop - Core i5 processor, 8GB DDR4 RAM, 1TB HDD (43), LCD Projector (01). Code blocks Node.js, Visual Code, Eclipse, Java, Jdk, Python	DMS 42 hrs/we	Shrinath Kattimani	Asst. Instructor	Diploma in ECE
2	Laboratory-02	60	Desktop - Core i3 processor, 8GB DDR3 RAM, 500GB HDD (60), LCD Projector (01) Code blocks Node.js, Visual Code, Eclipse, Java, Jdk, Python, Information	DMS/System S	Prabhavati Hiremath	Asst. Instructor	Diploma in CSE

3	Laboratory- 03	40	Desktop - Core i3 processor, 8GB DDR3 RAM, 500 GB HDD (44), LCD Projector (01) Code blocks Nodej.js, Visual Code Editor, Java IDE, Python, Oracle Cloud	Web Technolog	Ashwini Belligatti	Asst. Instructor	Diploma in ECE
4	Laboratory- 04	40	Desktop - Core i5 processor, 16GB DDR4 RAM, 512GB SSD HDD (42), LCD Projector (01) Code blocks Nodej.js, Visual Code Editor, Java IDE, Python	DBA 42 hrs/we	Laxmi Madiwalar	Asst. Instructor	Diploma in CSE
5	Laboratory- 05	40	Desktop - Core i3 processor, 8GB DDR4 RAM, 1TB HDD (42), LCD Projector (01), Code blocks Nodej.js, Visual Code Editor, Java IDE, Python	System Softwa	Sunita Benakanalamath	Instructor	Diploma in CSE
6	Laboratory -06	40	HP ProDesk 280 G2 MT CPU: Intel core i7 4770 3.40 GHz RAM: 8 GB DDR3 HDD: 500GB SATA Monitor: 19.5" LED LPD 1404 (04), Laptop ThinkPad X220 (04)	Computer Netw	Salma M.B.	Instructor	Diploma in ECE
7	Laboratory -07	40	Desktop - Core i7 processor, 8GB DDR3 RAM, 500GB HDD (36), Desktop - Core i3 processor, 8GB DDR3 RAM, 500GB HDD (08), LCD Projector (04), Code blocks Nodej.js, Visual Code Editor, Java IDE, Python	DMS /Web Tec	Praveenkumar Mane	Asst. Instructor	Diploma in CSE
8	Digital Electronics Lab	40	Desktop - Core i3 processor, 8GB DDR3 RAM, 500GB HDD: Quantity (03), Digital IC Trainer Kits PC-DT-DC (05), Digital IC Trainer Kits PC-DT-DC (05), Digital IC Trainer Kits PC-DT-DC (05)	Digital Electron	Shayeenbanu Jatiger	Asst. Instructors	Diploma in ECE
9	Computer Lab- 01	80	Desktop - Core i3 processor, 8GB DDR3, RAM, 500GB HDD (01), LCD Projector (01)	Algorithms \ L	B.C.Marigoudar	Asst. Instructor	Diploma in CSE
10	Computer Lab- 02	80	Desktop - Core i3 processor, 8GB DDR3 RAM, 500GB HDD (01), LCD Projector (01)	Algorithms Lab	Aishwarya Bare	Asst. Instructor	B Sc in CSE

D2. Safety Measures in Laboratories

Table No. D2.1: List of various safety measures in laboratories.

Sr. No	Laboratory Name	Safety Measures
1	Server room	• Access to the server room is restricted only to authorized personnel. • Regular data backup is maintained to prevent loss of important information. • UPS and power backup facilities are provided for uninterrupted server operation. • Fire extinguishers are installed near the server room for emergency situations. • Proper cable management is maintained to avoid accidental disconnection and hazards. • Server systems are regularly monitored and maintained for safe and efficient functioning.
2	Computer Laboratories	• Students are given separate login ID and password to save individual work in the server. • USB ports are disabled to avoid copying of files and information. • All computers and servers are electrically grounded. • Server room is provided with sufficient ventilation. • Students are instructed not to carry eatables inside the computer laboratory. • All systems are properly shut down after completion of laboratory sessions. • Students are advised to handle computers and peripherals carefully to prevent damage.
3	Electronic / Microcontroller Lab	• All electronic equipment and microcontroller kits are electrically grounded to avoid electric shock. • Students are instructed to switch off power supplies before making or changing circuit connections. • Proper handling of soldering irons and heated components is ensured to prevent burns and injuries. • Damaged wires, components, and connections are immediately replaced to avoid short circuits. • Fire extinguishers and first aid facilities are made available inside the laboratory. • Students are instructed to handle microcontroller boards, sensors, and electronic components carefully to prevent damage.

D3. Project Laboratory/Research Laboratory

List of project laboratory/research laboratory/Centre of Excellence.

S.N.	Name of the Laboratory
1	Centre for Artificial Intelligence Research(CAIR)
2	Centre of Excellence in Visual Intelligence (CEVI)

Centre for Artificial Intelligence Research (CAIR)



1. Objectives and Vision of CAIR

The Centre for AI Research (CAIR) at KLE Technological University, Hubballi, is an interdisciplinary research centre established to advance artificial intelligence research across three core domains: Intelligent Industrial automation and AI in Healthcare. The centre operates under the School of Computer Science and Engineering and is directed by Prof. Satyadhyan Chickerur.

CAIRs founding objectives are:

- To develop multi-objective optimized algorithms and models with a focus on system and architectural aspects of edge AI computing for vision-based applications in Industrial automation.
- To develop algorithms and software tools specifically designed for high-performance computing environments involving multi-core processors, accelerators (GPUs or FPGAs), and specialized HPC systems.
- To develop AI-based algorithms and systems for acquiring, integrating, and analysing medical information to extract knowledge for healthcare applications and research.

The centres proposed objectives have since been expanded to reflect its evolving research portfolio:

- Develop scalable and efficient algorithms and models that optimise system and architectural considerations for AI-based solutions across vision, speech, and text domains.
- Advance computational modelling, medical AI, genomics, drug discovery, and healthcare innovations through interdisciplinary solutions for global health challenges.

CAIRs vision is to serve as a hub where algorithms meet healthcare and intelligent automation — transforming images into insights and delivering scalable, real-time, intelligent solutions with measurable societal impact.

2. Research Focus Areas and Ongoing Activities

CAIR organises its research under three primary focus areas, each supported by dedicated faculty and research scholar groups.

2.1 Industrial Intelligent automation — AI/ML Model Optimization

The IIA group addresses redundancies in AI model parameters (neurons, weights, activations) to design optimised, lightweight, low-computational, and low-memory-bandwidth models for vision-based tasks. Key research directions include:

- Pruning: Adaptive and filter-based pruning techniques
- Quantization: Adaptive methods and normalization-based approaches
- LLM Optimization: Investigating redundancies in encoder-decoder architectures for image denoising and text-to-image applications; embeddings and attention matrix optimization
- Neural Architecture Search (NAS): Multi-objective optimized NAS for edge devices; search space design using genetic algorithm-based encoding optimization
- Explainable AI (XAI): Developing incremental learning frameworks for XAI in the agriculture domain; expandability, interpretability, and continuous learning

Student-level research directions under Edge AI include optimization of human activity recognition models, VGG-16 stripe-wise pruning with meta-learning, BERT optimization for low-latency applications, generative AI for heritage image enhancement, and image super-resolution via NAS.

2.2 AI in Healthcare - High Performance Computing (HPC) Research

The HPC group currently focuses on Computational Fluid Dynamics (CFD) and is expanding into Power-Aware Computing and Multispectral/Hyperspectral Imaging. Research activities include:

- Flow loop experimentation in cardiac models
- Computational modelling for haemodynamic using CFD
- CFD and physics-informed neural network (PINN) techniques for heart modelling
- AI-based prediction framework for stenosis changes in heart aorta models (collaborative, KIMS)

2.3 AI in Healthcare

This group addresses clinical AI applications through image analysis, image reconstruction, and predictive analytics. Current research directions for faculty and research scholars include:

- Deep learning-enabled coronary artery segmentation and plaque characterization for cardiac risk analysis
- Optimization of image reconstruction parameters and AI solutions for enhanced low-dose imaging
- XAI-driven image classification integrating heterogeneous data and biomarker identification
- Interpretability and Explainability models for Brain MRI classification
- Predictive modelling using multimodal data for chronic disease risk factor identification

2.4 CAIR–JNMC Collaborative Research Programme (AI in Healthcare)

A formal collaborative research programme between CAIR (KLETech, Hubballi) and Jawaharlal Nehru Medical College (JNMC, Belagavi) is actively underway. The second joint faculty meeting was held on 12 March 2026 at JNMC, Belagavi, co-chaired by Dr. Sulakshana Baliga (JNMC) and Prof. Satyadhyam Chickerur (KLETech). Eleven AI research projects have been formally identified and approved, with paired JNMC–KLETech faculty teams for each.

Key operational decisions include adoption of DICOM images and FHIR/HL7 standards, a standardised data collection process using direct capture and departmental repositories, and a formal AI protocol to be shared across all technical teams. A JNMC Principals Circular dated 7 April 2026 authorises data sharing with KLETech subject to permissions from the Principal, Medical Director, and the Institutional Ethics Committee.

2.5 CAIR–CARR Collaborative Research (AI in Intelligent Industrial Automation)

CAIR is engaged in a formal collaborative research programme with the Centre for Automation and Robotics Research (CARR) at KLETech. Key outcomes include confirmation that company-sponsored solution development may be aligned with the CAIR–CARR research track, and the assignment of one PhD candidate each from CAIR and CARR to the collaborative research programme.

3. Industry Collaborations and MoUs

CAIR maintains active collaborations with both industry partners and higher education institutions. The following table summarises the confirmed industry and HEI collaborations.

Partner / Institution	Nature of Collaboration
Microfinish	AI-Based Casting Defect and Document Discrepancy Detection
DANA India (Spicer)	AI-Driven Smart Industrial Automation and Real-Time Process Monitoring System
Magna International	AI-Driven Hardware-Agnostic Edge Intelligence and Industrial Vision Inspection Framework
JNMC, Belagavi	AI in Healthcare — 11 collaborative clinical AI research projects
CARR, KLETech	AI in Intelligent Industrial Automation — joint PhD-level collaborative research
Cardiff Medical School (Indo-UK – IIITD)	Collaborative research
KMC & RI	Collaborative research
SDM Dental College	Collaborative research
DIMS / DIMHANS	Healthcare AI collaboration
UAS (University of Agricultural Sciences)	Collaborative research

The three active industry-partnered projects under the AI in Automation track are described below:

AI-Based Casting Defect and Document Discrepancy Detection (Microfinish): AI models for automatic casting defect identification and document discrepancy detection in manufacturing workflows, aimed at reducing quality control time and improving process reliability at scale.

AI-Driven Smart Industrial Automation and Real-Time Process Monitoring System (DANA India): An integrated AI platform for real-time monitoring, predictive maintenance, anomaly detection, and data-driven process optimisation.

AI-Driven Hardware-Agnostic Edge Intelligence and Industrial Vision Inspection Framework (Magna International): A hardware-agnostic edge AI framework enabling AI/ML deployment across diverse hardware accelerators (GPU, DSP, NPU) for accurate and scalable quality inspection.

4. Student and faculty engagement details

4.1 Faculty and Research Scholar Composition

CAIR operates with a structured research hierarchy: Senior Research Group (SRG), Early Research Group (ERG), External Research Scholars (ERS), and Industry/Research Engagement Fellows (IREF). The current team composition is as follows.

Research Focus	Faculty / Staff Members
Intelligent Industrial Automation	Dr. Meena S.M, Dr. Uday Kulkarni, Dr. S R Nirmala, Dr. Sharada S, Dr. Praveen M.D, Sunil V Gurlahosur, Priyadarshini P, Satish Chickmath, Channabasappa M, Anand L (CARR), Shiva Prasad (CARR), Jai Venkat (MS-Research)
AI in Healthcare	Dr. Satyadhyan Chickerur, Dr. Shivashankar H, Dr. Guruprasad K, Dr. Sumaiya Pathan, Indira Bidari, Uma Hiremath, Preeti Y R Dr. S R Nirmala, Dr. Suvarna K, Dr. Shrinivas Desai, Prashant N, Kavita Chachadi, Nirmala Patil, Neha Tarannum, Nagaratna Y, Vijayalakshmi S, Shwetha P

4.2 Student Engagement

Category	Current Count
PhD Research Scholars	10
Postgraduate (PG) Students	10
Undergraduate (UG) Students	190

UG and PG students engage in research through the REU (Research Experience for Undergraduates) programme. Recent REU contributions include work in CFD-PINN for cardiac modelling, image reconstruction optimization, and edge AI model optimization.

4.3 CAIR–JNMC Faculty Pairs (Healthcare Projects)

Eleven faculty pairs from JNMC (clinical domain experts) and KLETech (AI/technical leads) are working on the approved healthcare AI projects. The JNMC faculty involved include Professors and HODs from Microbiology, Ophthalmology, Interventional Radiology, ENT, CTVS, Surgical Oncology, Skin and V.D., Pharmacology, and Anaesthesiology departments. KLETech faculty leads include Prof. Satyadhyan Chickerur, V S Hombalimath, Dattaprasad Torse, Srinivas Desai, Sujatha C, Swati Mavinkattimath, Nirmala S R, Santosh Pattar, Suvarna K, Prema Akkasaligar, Sujata Patil, Priyanka G, Uma M, Salma Shahapur, Guru K, and Rajashri Khanai.

5. Projects, Publications, Patents, Internships, and Funded Research

5.1 Approved Healthcare Research Projects (CAIR–JNMC)

The following eleven projects have been approved and are currently in progress under the CAIR–JNMC collaborative framework:

Sl.	Project Title	JNMC Lead	KLETech Lead(s)
1	Micromitra: A Chatbot for Microbiology Course	Dr. Sheetal Harakuni	Srinivas Desai, V S Hombalimath
2	AI-Based Dynamic Local Antibigram and Antimicrobial Resistance Trend Prediction System	Dr. Preeti S Maste	V S Hombalimath, Dattaprasad Torse
3	AI-Powered Portable Vein Visualization Device based on NIR Imaging for Lab and Bedside Phlebotomy	Dr. Chetana P. Hadimani	Satyadhyan Chickerur, R.H. Havaladar
4	AI Platform for Automated CEAP Classification and VCSS Scoring of Lower Limb Varicose Veins using Photographic Imaging and Ultrasound Doppler Analysis	Dr. Iranna Hittalmani	Sujatha C, Swati Mavinkattimath
5	AI-Powered mHealth Solution for Infant Hearing Screening in Low-Resource Settings	Dr. Priti Hajare	Nirmala S R, Santosh Pattar
6	AI-Based Dosage Prediction Model for Vitamin K Antagonists using Serial PT/INR Values in Post-Valve Replacement Patients	Dr. Ranjit Naik	Suvarna K, Prema Akkasaligar
7	Role of AI in the Diagnosis, Treatment, and Management of Oral Cavity Cancers	Dr. Kumar Vinchurkar	Sujata Patil, Priyanka G
8	AI in Quality Control of Suturing among Undergraduates and Dermatology Residents	Dr. Bhavana Doshi (Dr. Shiva)	Uma M, Salma Shahapur
9	AI-Based Prediction of Self-Medication Risk among Medical Students	Dr. Jyoti M. Benni	Srinivas Desai
10	Artificial Intelligence in Airway Anaesthesia	Dr. Rajesh Mane	Satyadhyan Chickerur, Guru K, Rajashri Khanai
11	Blockchain-Enabled De-Identified Digital Health Asset Repository Supporting Federated Learning	Dr. Iranna Hittalmani	Satyadhyan Chickerur

5.2 Publications and Patents

The following table summarises the publication and patent output of CAIR, comparing current performance against targets.

Year	Journal Publications	Conference Publications	Patents Filed / Granted
2025	06	48	02
2024	05	50	-
2023	02	39	-

5.3 Fundings

CAIR has applied for 5 funding grants and has identified 15 additional grants to be pursued in the next cycle. The current funding KPI stands at 10% achievement, with a target of 30%. Funding avenues being explored include national schemes under DST, MeiT, and SERB, as well as industry-sponsored research projects.

5.4 Industry Projects

A) AI-Based Casting Defect and Document Discrepancy Detection

Partner: Microfinish

AI models for automatic casting defect identification and document discrepancy detection in manufacturing workflows, reducing quality control time and improving process reliability at scale.



B) AI-Driven Smart Industrial Automation and Real-Time Process Monitoring System

Partner: DANA India

Integrated AI platform for real-time monitoring, predictive maintenance, anomaly detection, and data-driven process optimisation.



C) AI-Driven Hardware-Agnostic Edge Intelligence and Industrial Vision Inspection Framework

Partner: Magna International

Hardware-agnostic edge AI framework enabling AI/ML deployment across diverse hardware accelerators (GPU, DSP, NPU) for accurate and scalable quality inspection.



6. Workshops, Technical Events, and Training Programs

A) MED-AI Hackathon — Innovate. Collaborate. Impact.

May 2026 | KLE Technological University, Hubballi

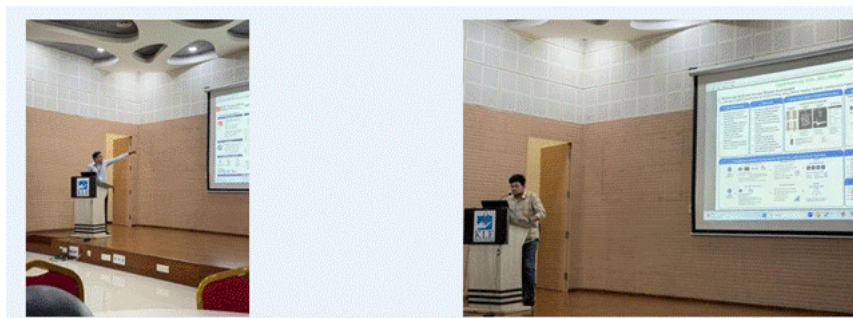
CAIR organised and hosted the MED-AI Hackathon, bringing together students and researchers to ideate and prototype AI-driven solutions for healthcare challenges. The event saw the participation of 20 teams (20 students from JNMC, Belagavi / JGMMMC, Hubballi,) who worked alongside KLETech students to develop innovative solutions spanning medical imaging, clinical decision support, and patient monitoring. Teams worked on problem statements aligned to the active CAIR–JNMC research projects, reinforcing the collaboration model between engineering and medical institutions.



B) Student e-Poster Presentations — Digital Health Day 2026

2026 | KLE Centenary Convention Centre, JNMC Campus, Belagavi

Two student researchers (Mr. Keshava Mahaek K H – CAIR) and (Mr. Shashank Joshi – CEVI) presented e-posters at the Digital Health Day 2026 symposium held at JNMC, Belagavi. The presentations reflected work carried out under the CAIR/CEVI–JNMC collaborative research programme and covered AI-driven clinical applications including multimodal AI for venous disease assessment and neuroscience-based signal analysis. The presentations demonstrated the research maturity of CAIR/CEVI -affiliated students and marked a significant milestone for UG-level AI research at KLETech.



C) AI Workshop — AROKSCON 2026 National Radiology Conference

Saturday, 16 May 2026 | KLE Centenary Convention Centre, JNMC Campus, Belagavi

Prof. Satyadhyan Chickerur, Director, CAIR, was invited as a speaker and workshop facilitator at AROKSCON 2026 — the national conference of the Association of Radiologists of Karnataka. The workshop, titled “AI Workshop: Algorithms to Action,” included a session on Talking to AI: Practical Prompt Engineering for Clinicians (8:50 AM). Prof. Chickerur addressed a packed auditorium of radiology professionals and clinicians, covering hands-on AI concepts for clinical workflows. The session generated significant interest in AI–radiology collaboration and CAIR’s research agenda.



6.3 AI Workshop — AROKSCON 2026 National Radiology Conference

Prof. Satyadhyan Chickerur, Director, CAIR, was invited as a speaker and workshop facilitator at AROKSCON 2026 — the national conference of the Association of Radiologists of Karnataka — held on Saturday, 16 May 2026 at the KLE Centenary Convention Centre, JNMC Campus, Belagavi. The workshop, titled AI Workshop: Algorithms to Action, included a session on Practical Prompt Engineering for Clinicians at 8:50 AM. Prof. Chickerur addressed a packed auditorium of radiology professionals, covering hands-on AI concepts applicable to clinical workflows. The session generated significant interest in AI–radiology collaboration and CAIRs broader research agenda.

6.4 GPU Programming and AI Workshops (Prior Cycles)

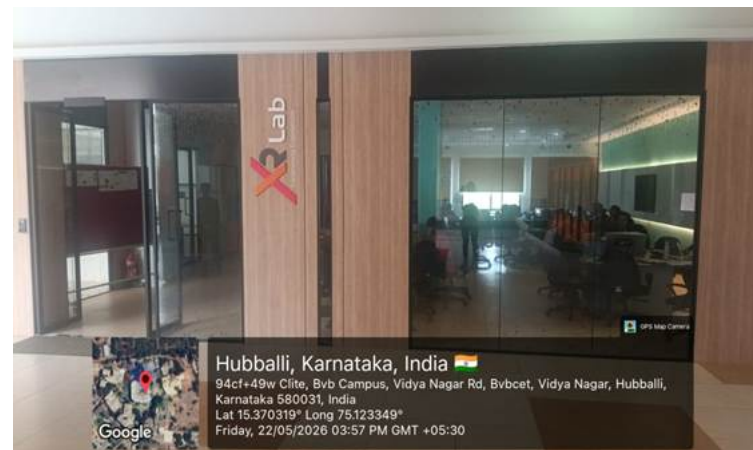
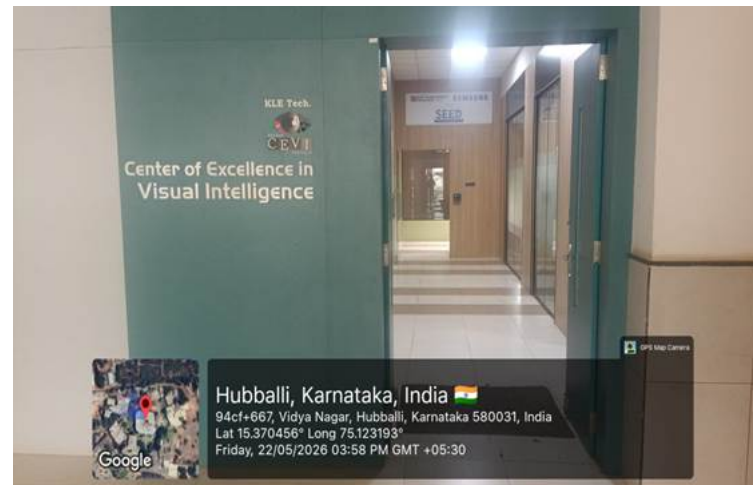
CAIR has previously conducted multi-day faculty development programmes (FDPs) and workshops in GPU programming and AI, including AISSMS FDP and similar programmes covering PyTorch, TensorFlow, CNNs, semantic segmentation, 3D point cloud processing, computational chemistry with RDKit, model optimisation, and deployment. These programmes have been delivered across multiple institutions.

7. Achievements, Recognitions, and Notable Outcomes

7.1 Institutional and Collaborative Milestones

- Eleven AI research projects formally approved and launched under the CAIR–JNMC collaborative programme — a landmark for interdisciplinary clinical AI research at KLETech.
- Official JNMC Principals Circular (7 April 2026) authorising data sharing and confirming AI algorithm development responsibility to the CAIR Laboratory — a formal institutional endorsement.
- Three active industry-sponsored projects executing under CAIR–CARR collaborative automation track, with partners Microfinish, DANA India, and Magna International.
- CAIRs collaborative network now spans 13 institutions and companies, including IIT Palakkad, Cardiff Medical School (Indo-UK), and multiple regional hospitals and universities.

Centre of Excellence in Visual Intelligence (CEVI)



Centre for Visual Intelligence (CEVI) at KLE Tech focuses on providing visual intelligence-based solutions to real-world problems. Research involves generative tasks like restoration in different scenarios, synthetic image generation, and discriminative tasks like classification, object detection/ tracking, & semantic segmentation on both 2D and 3D data. CEVI focuses on the development of new computational methods and their innovative applications to interpret the problems in the areas of Digital Heritage, Agriculture, and Smart Manufacturing and also demonstrates/ presentations through Extended Reality.

1: Objectives

- To facilitate institute-industry collaborations and form advisory groups.
- To foster interdisciplinary research towards collaborations with leading international research centers working towards socio-technical goals.
- To develop products through special interest groups and elevate to commercialization by setting up start-ups.

2: Infrastructure and facilities available

CEVI Lab: 3028 Sqft

- Dark Room(30x25): 750 sqft
- Green Room(18x20): 360 sqft
- Motion Capture(20x20):400 sqft
- Server Room(10x12): 120sqft
- Discussion Area(58x12):696 sqft
- SEED(30x32): 960 sqft

XR Lab: 1792 Sqft

- AR/VR/XR Lab (22x22): 484 sqft
- Processing Room-01(08x22): 176sqft
- Processing Room-02(18x10): 180sqft

3: (a) Hardware and software resources (Internal Funding) :

- Imatest Equipement for Darkroom
- HP 280 G6 MT Systems: 20 systems 16Gb RAM 1TB HDD Intel Core i7-10700
- Staging Server
- Microsoft HoloLens 2
- Camera: EOS R5 and EOS R6
- Mini Handloom Machine.
- Work Stations: 02(2X NVIDIA RTX 3090)
- Imatest Equipement(Software)

3: (b) Hardware and software resources (External Funding):

- MetaQuest3 :02
- Work Stations: 06
- NAS(40TB) : 01
- Mac mini
- I PAD pro
- Mac Books: 03
- Orbbec Femto Bolt camera: 08
- Apple Developer Account(Software)

4: Research focus areas and ongoing activities

2D Data Acquisition / Capture Setup

- 2D Data Acquisition and Curation
- Restoration and Enhancement of Images
- Categorization, Clustering and Classification
- Object Detection and Tracking
- Low light Image capture with varied lighting conditions.

Learning Representation of 2D Data

- Transforming 2D data into formats suitable for utilization
- Efficient analysis and interpretation of this information
- image classification, object recognition, and image segmentation

Image Restoration and Enhancement

- Underwater Image Restoration and Enhancement
- Physics based lowlight Image Enhancement
- Image Denoising, Deblurring, Dehazing, Color normalization.

Filtering, Classification, and Categorization of 2D Data

- Heritage image classification/categorization towards 3D reconstruction

Quality Analysis of Images

- Image quality assessment (reference-based and non-reference based)

Detection, Localization, and Tracking

- Detection, Localization and Tracking
- Detection & localization of oral cancer tissues in histopathological slide images
- Tracking of tools during surgery towards automated evaluation of trainee neurosurgeons

Engineered data (SEED)

- Data Capture Techniques
- Automation/ Semi Automation of Annotation
- Synthetic Data Generation
- Parameters for Quality Analysis.

5(a): Industry collaborations and MoUs

- Samsung Research Institution Bangalore(SRI-B): MOU till 2028
- Docketrun

5(b): Academia collaborations

- AIIMS Delhi
- IIT Delhi
- KLE VKIDS
- IIIT Hyderabad

6: Student and faculty engagement details

	2023-2024		2024-2025		2025-2026	
	Projects	Students	Projects	Students	Projects	Students
REU	10	10	5	5	2	2
SRP	5	10	5	11	0	0
Mini	23	83	23	88	15	59
Minor	23	83	23	88	15	59
SDP	8	23	10	35	20	77

Internship			5	5	7	7
Total Students Engaged		209		232	-	204

7: Projects, publications, patents, internships, and funded research supported by the lab.

Projects	Nos
Ongoing Projects	01
Completed Projects	10
Patents filed:	05
Internship:	07

Ongoing Projects

1. SERB-INAE KALPIT(2024-2026)

Completed Projects

1. DST IHDS-Crowdsourcing 2019-2024
2. DST Digital Poompuhar 2019-2024
3. VGST V-Vaidya 2021-2023
4. AICTE-RPS Shape Representation 2019-2023
5. DST: India Digital heritage 'Acquisition, representation and rendering of digital heritage sites' in collaboration with IIT Delhi
6. DST: India Digital heritage 'Gesture based virtual walkthrough of digital heritage sites
Hampi' (2011-2015)
7. IEEE EPIC scheme: Communicate 'A Hand Speech Glove (2010-2018)
8. Naval Research Board (NRB): 'Super-resolution and retargeting in Images and Videos' (2009-2012) in collaboration with IIT Delhi and IIIT Hyderabad
9. DRDO, ADE- 'Robust horizon detection for UAV' (2013 - 2014)
10. VTU: 'Image Fusion and Quality metrics for images' (2010-2014)

Publications

Details	2022	2023	2024	2025	2026 (Till Date)
Q1	10	8	9	8	15
Q2	0	0	0	2	0
Q3	6	7	2	6	0
Q4	6	3	5	6	0

NA	0	0	8	2	1
Total	22	18	24	16	16

8: Workshops, technical events, and training programs conducted

- CEVI Visual Intelligence Summer School workshops for KLETech Students - every year, 3 weeks workshop
- Conducted First Edition of Marine Vision Restoration Challenge (MVRC) at 3rd Workshop on Maritime Computer Vision Workshop (MaCVi), at IEEE/CVF Winter Conference on Applications of Computer Vision 2025.

9: Any achievements, recognitions, or notable outcomes

- Selected amongst top teams in NTIRE Challenges on Image Restoration and Enhancement @ CVPR 2022,2024,2025 and 2026 and published 25+ joint reports (top 3 in 3 challenges)
- Selected amongst top teams @ MIPI CVPR 2023 Published a Joint Report on Flare Removal
- Selected amongst top teams @ MaCVi WACV 2023. Published a Joint Report on Detection of Objects in Sea
- Detection of Drone in Wild (CCTV Feeds)- by DRDO ; Rs 1,00,000 Cash Prize
- Reconstruction and Presentation of Cultural Heritage by Amazon ; Rs 50,000 Cash Prize
- Samsung Best Partner ACE Award 2022 to KLETech
- 16 Teams Awarded Excellence Award by PRISM In two Years (\$900 Worth Rewards/ Team)
- Travel Grants to Faculty and Students

i: 3 Grants for ICCV 2023

ii: 2 Grants for AAAI 2024

- Student volunteering at SIGGRAPH Asia at Tokyo 2024.

PART E: First Year faculty and financial Resources

(Data to be filled in for the first year course faculty and budget allocation and utilization)

E1. First Year Student-Faculty Ratio (FYSFR)

Table No. E1.1: FYSFR details.

Year	Sanctioned intake of all UG programs (S4)	No. of required faculty (RF4= S4/20)	No. of faculty members in Basic Science Courses & Humanities and Social Sciences including Management courses (NS1)	No. of faculty members in Engineering Science Courses (NS2)	Percentage= No. of faculty members ((NS1*0.8) + (NS2*0.2))/(No. of required faculty (RF4)); Percentage= ((NS1*0.8) +(NS2*0.2))/RF
2023-24(CAYm2)	1380	69	46	22	60
2024-25(CAYm1)	1380	69	49	27	63
2025-26(CAY)	1800	90	57	27	54

E2. Budget Allocation, Utilization, and Public Accounting at Institute Level

Table No. E2.1: Budget and actual expenditure incurred at Institute level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Infrastructure Built-Up	275000000.00	266400332.26	292200000.00	337836058.50	320300000.00	275490296.08	226000000.00	272900679.00

Library	10800000.00	11640291.79	11600000.00	9771428.06	8100000.00	6934858.90	6550000.00	6684290.00
Laboratory equipment	77000000.00	70298745.00	40000000.00	42923569.00	30000000.00	42637137.00	60000000.00	45115063.00
Teaching and non-teaching staff salary	583700000.00	573446030.00	561500000.00	551937567.00	512840000.00	498123229.00	508167243.00	486223533.00
Outreach Programs	2000000.00	1821446.00	1700000.00	1059445.00	1200000.00	1185057.00	1800000.00	1759579.55
R&D	95000000.00	89863171.00	120000000.00	105715282.20	91000000.00	67685714.98	60500000.00	85916358.00
Training, Placement and Industry linkage	40000000.00	32995616.20	28000000.00	28096690.28	39400000.00	27868861.85	16700000.00	22804772.45
SDGs	500000.00	00	500000.00	00	00	00	00	00
Entrepreneurship	6000000.00	5500402.00	6500000.00	5765247.00	8500000.00	8288555.00	6000000.00	5721313.00
Others, specify	318247000.00	300391085.97	240450000.00	236975209.38	221575000.00	222576706.76	200300000.00	212374205.00
Total	1408247000.00	1352357120.22	1302450000.00	1320080496.42	1232915000.00	1150790416.57	1086017243.00	1139499793.00

E3. Budget Allocation, Utilization, and Public Accounting at Program Specific Level

Table No. E3.1: Budget and actual expenditure incurred at program level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Laboratory equipment	3590000.00	495102.00	2500000.00	2066618.00	1800000.00	1755518.00	4000000.00	3755243.00
Software	0	0	0	0	0	0	0	0
SDGs	50000.00	0	50000.00	0	0	0	0	0
Support for faculty development	1800000.00	1669506.25	1200000.00	768708.00	3000000.00	2579986.00	275000.00	262448.00
R & D	31500000.00	30197829.00	55000000.00	52521861.00	13750000.00	18445302.00	20000000.00	16935426.00
Industrial Training, Industry expert, Internship	4100000.00	3832726.25	3000000.00	3533512.00	2500000.00	2374707.00	4625000.00	4586680.00
Miscellaneous Expenses*	202355000.00	197212139.24	184300000.00	182171354.10	175515000.00	165465824.87	187115000.00	195192062.00
Total	243395000.00	233407302.74	246050000.00	241062053.10	196565000.00	190621337.87	216015000.00	220731859.00